

6(a)	volt / ampere	B1
6(b)(i)	$R_T = [1/3.0 + 1/6.0]^{-1} + 4.0 (= 6.0 \Omega)$	C1
	$I = 1.5 / 6.0$	C1
	$= 0.25 \text{ A}$	A1
6(b)(ii)	$V_B = 0.5 \text{ V}$ $I = 0.5 / 3.0$ $= 0.17 (0.167) \text{ A}$	A1
6(b)(iii)	$P = I^2 R \text{ or } VI \text{ or } V^2 / R$	C1
	$\text{ratio} = (0.167^2 \times 3.0) / (0.25^2 \times 4.0)$ $= 0.33$	A1
6(c)(i)	vary/change/different radius/diameter/ <u>cross-sectional</u> area (of wire)	B1
6(c)(ii)	$v = I / Ane$ $\text{ratio} = \frac{(I_B / A_B)}{(I_C / A_C)} \text{ or } \frac{I_B}{I_C} \times \frac{A_C}{A_B}$	C1
	$(R \propto 1 / A \text{ so}) \text{ ratio} = \frac{I_B}{I_C} \times \frac{R_B}{R_C} = \frac{0.167}{0.25} \times \frac{3.0}{4.0}$ $= 0.50$	A1
6(d)(i)	0.25 A to 0.13 (0.125) A or halved	A1
6(d)(ii)	no change	A1